37

i Mendel selected pea plants for his experiments as -

* ~~Thought~~ These plants showed various characteristics that were easily identifiable and helped in ~~the~~ getting accurate results. eg - Height, seed color etc

* Pea plant's flowers are bi-sexual, so ~~cross~~ self pollination and cross pollination could be done easily.

* Pea plants have very less maturing time and grow very fast. This allowed Mendel to perform experiments at a faster rate.

* Pea plant is easy to grow, and requires less maintainance. So, it could be grown easily by anybody ~~at~~ without any trouble and in numerous numbers.

ii Dominant traits are the traits which are shown or expressed in an individual ~~to~~ and supress the recessive trait

Recessive trait is the trait which is not shown and suppresed in an individual by the dominant trait.

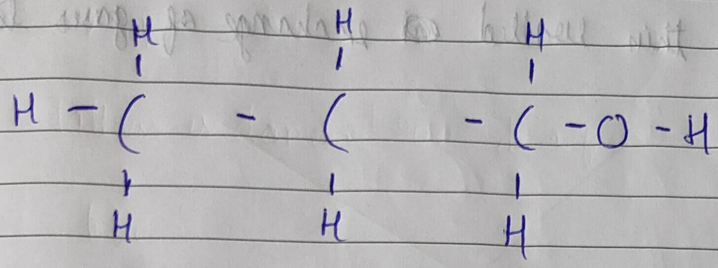
eg In pea plant, Tall height is a dominant trait but dwarf is a recessive trait

iii Seed Shape → Round (dominant), Wrinkled (recessive)

Seed color → Yellow (dominant), Green (recessive)

classmate PAGE

25 Propanol (C_3H_7OH)



26

i Aluminium reacts with water only in gaseous form (steam)
Magnesium reacts with hot water only

ii Copper does not react with steam

27 The lung is a complex organ in which exchange of gases take place. To ~~a~~ do this effisiently lungs ~~...~~ maximise the area for exchange of gases by

* The lungs are divided into bronchi which divide into bronchioles and then alveoli.

* Alveoli are small ~~...~~ baloon like structures that are richly supplied with blood ~~...~~ vessels.

* They help in increasing the surface area for exchange of gas

* Since they are thin walled exchange of gases takes place easily.

28

a Liver - Liver produces bile juice for emulsification of fat

Pancreas: Secrete insulin and glucagon to regulate blood sugar level

b Nutrition - The intake of food and nutrients by living organisms from the surrounding to sustain life processes is called nutrition.

Nutrients - The necessary minerals and substances present in food which are needed for living organisms to survive. eg - vitamins, proteins etc.

	Holozoic Nutrition		Saprophytic Nutrition
1	The digestion of food happens inside the body of organism	1	Digestion and breakdown of food happens outside the body
2	Involves complex living organic matter	2	Involves dead, decaying matter
eg	Tiger, human		eg - Bacteria, fungi

29 Self pollination - When one flower of a plant is pollinated by another flower ~~of~~ in the same plant, It is called self-pollination. Or if the bi-sexual flower ~~is pattern~~ is pollinated with itself, it ~~is called~~ is also called self pollination.

Cross pollination happens between flowers ~~of~~ belonging to different plants of same species.

Abiotic agents such as wind, water etc. and biotic agents such as butterflies, bees and other insects help in pollination.

30 ~~Heat~~ Heat (H) $\propto$ Current (I)

H $\propto$ Resistance (R)

H $\propto$ Time (t)

$H = I^2Rt$ (Joule's law of heating)

$R = \frac{\rho l}{A}$

$l = 12m$

$A = \pi r^2 = 3.14 \times (2 \times 10^{-4} m)^2$

$= \frac{3.14 \times 10^{-8} \times 12}{3.14 \times (2 \times 10^{-4})^2} = \frac{6 \times 10^{-8}}{10^{-4}}$

$= \frac{12 \times 10^{-8}}{4 \times 10^{-8}} = 3$

∴ Resistance = 3 Ω

31 ~~P = 0.5~~ d

i $P = -0.5\ d$, $P = \frac{1}{f}$

$-0.5 = \frac{1}{f}$, $\frac{-5}{10} = \frac{1}{f}$

$f = -2\ m$

Principal focus

∴ focal length of lens needs to be -2 m or^ $\bar{2}$ m in front of lens.

ii $P = +1.5\ d$, $P = \frac{1}{f}$

$1.5 = \frac{1}{f}$, $\frac{15}{10} = \frac{1}{f}$

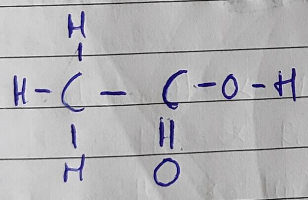
$f = \frac{\cancel{10}}{\cancel{15}} = \frac{2}{3}$ m or ~~0.66 0.67 m~~ 0.66 m (0.67 m approx)

∴ focal length of lens needs to be 0.66 m or principal focus 0.66 m behind lens.

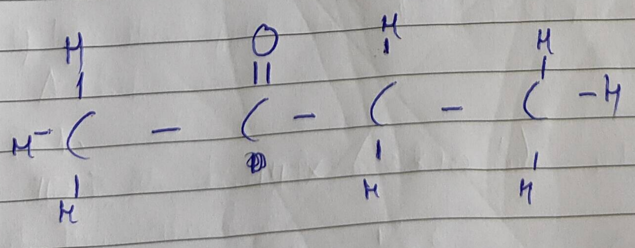
32 Metal - Mercury (Hg)
Ore - Cinnabar (HgS)

Roasting -
$2HgS + 3O_2 \rightarrow 2HgO + 2SO_2$
~~$HgO + HgS \rightarrow Hg + SO_2$~~ $HgS + HgO \rightarrow Hg + SO_2$

33
i Ethanoic acid (CH_3COOH)



ii Butanone (C



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DATE

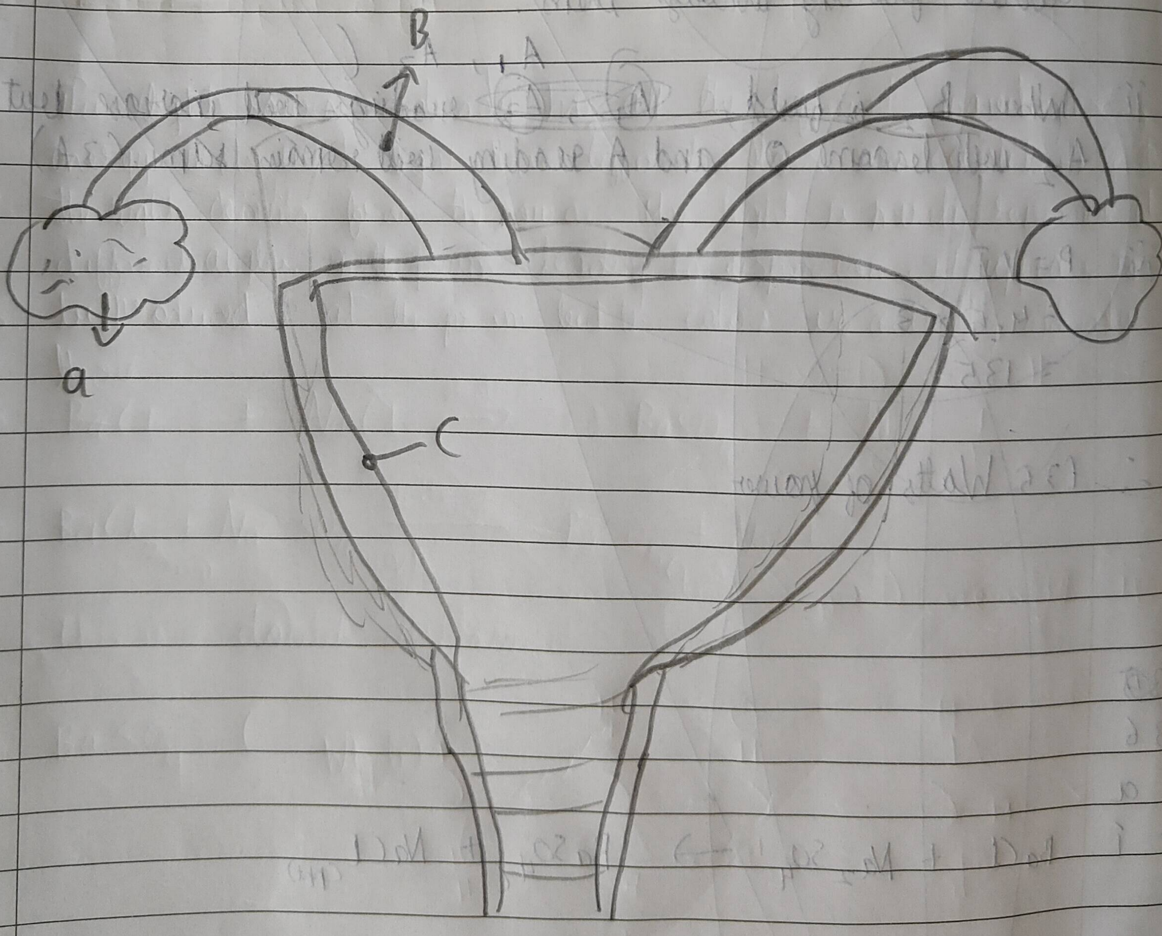
11

34
i
a Ovary

b ~~Uterus~~ Fallopian Tube

c Uterus (wall of uterus)

ii ~~H~~ Human egg if not fertilised, does not mature. ~~It dies~~ It implants itself in the uterin wall which is later shed off and comes out of the body as mucous and blood.



(3)

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34

i $2NaCl_{(aq)} + 2H_2O_{(l)} \xrightarrow{electricity} 2NaOH_{(l)} + Cl_{2(g)} + H_{2(g)}$

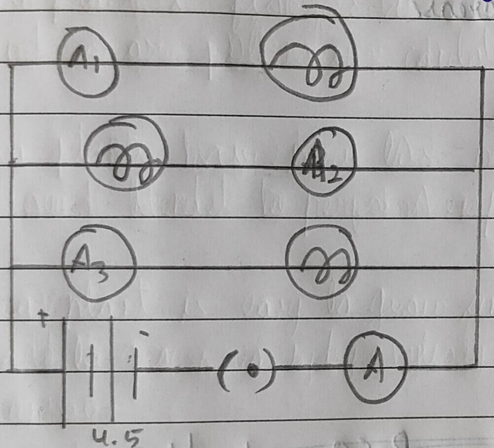
ii Anode - Cl_2 (Chlorine gas

Cathode - H_2 (hydrogen gas)

iii Chlorine gas - Used in disinfection of water and preparation of PVC

Hydrogen gas - Used in making ammonia and fuels.

35



Let resistance be R, of each bulb

$\frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{1}{R_{net}}$, $R_{net} = \frac{R}{3}$

$V = IR$, $4.5 = \frac{3 \times R}{3}$, $R_{net} = 4.5\,\Omega$

1.5

∴ Each bulb has resistance of ~~[illegible]~~ Ω

Since bulbs are identical 1 A will flow through each of them.

ii If B_1 is fused

$V = IR$,

$4.5 = I \times \frac{\cancel{1.5}}{2}\ \frac{4.5}{2}$

$I = 2\,A$ in whole circuit

1

Full Test -01

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1 4: b and d

2 3: Oxygen

3 1: Tallness is dominant over dwarfness

4 3: b, c

5 3: Translocation

6 1: Arteries

7 2: A convex mirror

8 3

9 1: Scattering of light is not enough at such heights

10 2: Refraction, dispersion and total internal reflection

11 ~~1: Low re~~ 2: High resistance, low melting point

12 2

13 3 None of the above

14 1: an exothermic reaction

15 1: CO_2

16 2: Z

17 3: i and iii

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18 3: 1-c, 2-d, 3-b, 4-a

19 2

20 2:148

21 Plants need raw materials like water, carbon-dioxide to carry out photosynthesis. It gets water from the soil which is absorbed by the roots and CO_2 from atmosphere takes in by stomata.

22 Deficiency of haemoglobin would lead to serious consequences. Haemoglobin is a pigmented protein responsible for transport of oxygen throughout the body as oxygen has high affinity towards it. Deficiency of haemoglobin would result in the transport of oxygen becoming inefficient and thus may not be able to meet our body's oxygen needs leading to hypoxia. Also the color of blood may change.

23 The current carrying conductor will experience no force when it is placed parallel to the direction of magnetic field and the angle between the direction of current and direction of magnetic field is 0°.

24 The image formed by plane mirrors is -

* Height of image is equal to height of object. The magnification is +1.

* The image is virtual, erect and laterally inverted and same size.

* The distance between the image and mirror is equal to the distance between mirror and object. (u=v)

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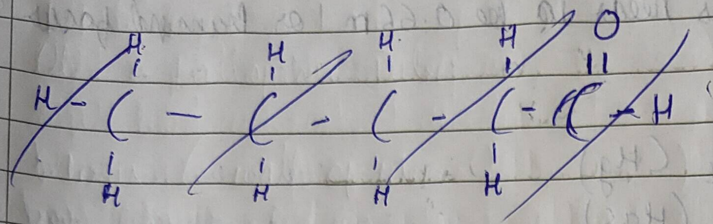
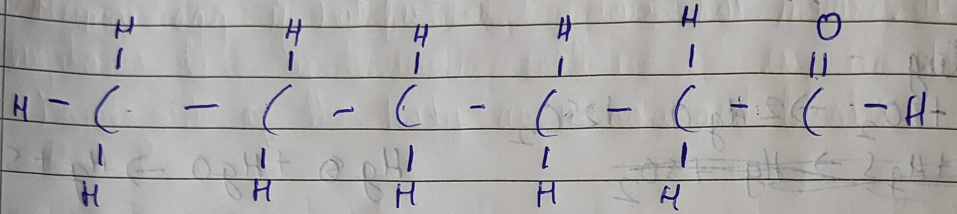
iii Hexanal ($C_5H_{11}CHO$)





34 ii ~~If the human egg is not fertilized it will~~

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38
a 380 nm to 780 nm

c ~~5~~ White light is made up of 7 different colors. When ~~they~~ white light is passed through a prism, the ~~colors~~ different colors of light bend and refract at different angles. This is because they have different properties (wavelength) and thus dispersion of light occurs. Red light bends the least while violet bends the most. Different colors travel at different speeds

~~d~~

~~e~~ ~~Bif Bit~~

6 The light source already has all the different wavelengths of energy in its white light. These different colors are constituents of the the light coming from the source. The light is merely refracted at different angles in the prism leading to dispersion. It appears to be produced ~~d~~ at the prism but the colors are actually already there.

We know $P \propto I$ in parallel circuit, so I increases, The other bulbs will glow brighter

ii If B_2 is fused

~~$A_1 = 1A$~~ $A_1 = 1A$
~~$A_2 = 0A$~~ $A_2 = 0A$
~~$A_3 = 1A$~~ $A_3 = 1A$
~~$A = 2A$~~ $A = 2A$

A_1, A_3 reading will increase but A_2 will become 0 A and A will ~~remain same~~ decrease

iii $P = VI$
$= 4.5 \times 3$
$\therefore = 13.5$ Watts

36
a

~~$BaCl + NaCl \xrightarrow{electricity} BaSO_4 + Na$~~

i $BaCl_{2(aq)} + Na_2SO_{4(aq)} \xrightarrow{electricity} BaSO_{4(ppt)} + NaCl_{(aq)}$

ii This is a double displacement

iii $BaSO_4$ (White)

b Slaked lime ($Ca(OH)_2$) is substance used for white washing of walls. On drying, $CaCO_3$ is produced which gives shiny finish to walls.